

Apply for a Science Corps Fellowship - Entry #3237

Name

Zheng Tong

Email

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Gender

Male

City, state, and country of residence

Oxford

Nationality

China

When did or will you finish your PhD?

June 2024

University Education

2016-2020, BEng in Mechanical Engineering with Renewable Energy (First class degree with Hons), School of Engineering, The University of Edinburgh.

2020-2024, DPhil in Engineering Science, Harris Manchester College, University of Oxford. Supervisor: Prof. Stephen J. Payne

Thesis title: The Full-brain 3D Computational Modelling of Cerebral Autoregulation Based on Haemodynamics.

What are your reasons for applying to be a Science Corps fellow?

I am applying to be a Science Corps fellow because I believe in the transformative power of STEM education and its potential to uplift underserved communities worldwide. My background in Mechanical Engineering and my current research in cerebral autoregulation have equipped me with a strong foundation in STEM, and I am passionate about sharing this knowledge with others.

Science Corps offers a unique opportunity for STEM Ph.D. graduates like me to step outside the traditional academic path and make a tangible impact on education in regions that lack access to quality science teaching. The organization's mission aligns with my desire to broaden my perspective and skill set while contributing my expertise to underserved communities.

My experience in building computational models and conducting research has honed my problem-solving skills and ability to communicate complex ideas effectively. As a Science Corps fellow, I would use these skills to design a scientific curriculum, teach, host workshops, and conduct experiments, providing students with hands-on STEM experiences that can inspire and empower them to pursue careers in STEM fields.

In essence, I see Science Corps as a bridge between my academic background and my aspiration to make a meaningful difference in the world through STEM education.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

In addition to my academic pursuits, I have accumulated significant experience that has well-prepared me for a Science Corps fellowship. My background includes tutoring high school students in A-level courses and developing curriculum designs for educational institutions. I have also undertaken the task of designing and implementing research-based courses.

Notably, I've assisted over 30 students in successfully securing admissions to prestigious G5 schools, which highlights my proficiency in guiding and mentoring students through rigorous academic pathways. My ability to communicate effectively with students and nurture their curiosity and interest in science has been a defining aspect of my teaching approach.

Moreover, my role in curriculum design and research course development has honed my skills in creating structured, engaging, and effective learning materials. This experience will be invaluable in crafting a scientific curriculum as a Science Corps fellow.

Combining my academic achievements in STEM with my extensive experience in teaching and mentoring, I am well-equipped to excel as a Science Corps fellow. I am enthusiastic about applying my skills to inspire and empower students in underserved communities, fostering a love for STEM subjects, and helping them pursue educational and career opportunities in these fields.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Doing science in industry

In science education

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

In the near future, within the next two years, I envision myself continuing my academic journey by completing my DPhil in Engineering Science. During this time, I aim to publish my research findings and contribute to the academic community's understanding of cerebral autoregulation. I also plan to engage in further teaching and mentoring experiences, possibly as a teaching assistant or research assistant at my university, to continue honing my pedagogical skills.

In the long term, spanning the next ten years, my professional aspirations revolve around a career in education and academia. I see myself as a university professor specializing in biomedical engineering or a related field. I aim to establish my research lab dedicated to studying cerebral autoregulation and its clinical applications. Simultaneously, I want to be actively involved in curriculum development, creating innovative and inclusive STEM courses that inspire students to explore the wonders of science and engineering. Ultimately, my goal is to foster the next generation of STEM leaders and contribute to advancements in both research and education in my chosen field.

CV/Resume

 [CV-Zheng-Tong-University-of-Oxford.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

Empty

[Science Corps](#)